WeightedCorrelations_H2A-MHC

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```
library(tidyverse)
library(readxl)
library(knitr)
# Data:
H2A <- read_csv("~/internship/workspace/Written Datasets/H2AWorkers_County_Clean_2.csv") # Worker densi
MHC <- read_csv("~/internship/workspace/Data/migrant_health_centers_ncfh.csv") # Migrant health centers
ENV <- read_csv("~/internship/workspace/Written Datasets/FEMA_Disasters_NumEventsGreaterThanZero.csv")
AGO <- read_csv("~/internship/workspace/Written Datasets/AgrCensus_Output_Clean.csv") # Agricultural ou
JUL23 <- read_csv("~/internship/HICAHS/Data/Heat_Ag_HumanRisk/CountyMaxTemp_JUL23.csv") # Maximum tempe
AUG23 <- read_csv("~/internship/HICAHS/Data/Heat_Ag_HumanRisk/CountyMaxTemp_AUG23.csv") # Maximum tempe

# Healthcare facilities in each state
# (Cannot bind due to differing number of columns)
COH <- read_excel("~/internship/workspace/Health Facility Data/ColoradoHealth24_xlsx.xlsx", sheet = 2)
WYH <- read_excel("~/internship/workspace/Health Facility Data/WyomingHealth24_xlsx.xlsx", sheet = 2)
SDH <- read_excel("~/internship/workspace/Health Facility Data/SouthDakotaHealth24_xlsx.xlsx", sheet =
NDH <- read_excel("~/internship/workspace/Health Facility Data/NorthDakotaHealth24_xlsx.xlsx", sheet =
UTH <- read_excel("~/internship/workspace/Health Facility Data/UtahHealth24_xlsx.xlsx", sheet = 2)
MTH <- read_excel("~/internship/workspace/Health Facility Data/MontanaHealth24_xlsx.xlsx", sheet = 1)


# Empty vectors for later
response <- character()
predictor <- character()

# Cleaning Cleaning Cleaning
ENV$county[ENV$county == "lamoure"] <- "la moure"
ENV <- ENV |> select(-matches("coastal|hurricane|tsunami|volcanic", ignore.case = TRUE))
AGO[] <- lapply(AGO, function(col) gsub(",", "", col))

WYH$county <- gsub(" County", "", WYH$county)
MTH$county <- str_to_title(MTH$county)

COH$state <- str_to_title(COH$state)
WYH$state <- str_to_title(WYH$state)
SDH$state <- str_to_title(SDH$state)
NDH$state <- str_to_title(NDH$state)
UTH$state <- str_to_title(UTH$state)
MTH$state <- str_to_title(MTH$state)

JUL23$Name <- gsub(" County", "", JUL23$Name)
AUG23$Name <- gsub(" County", "", AUG23$Name)
```

```r
JUL23 <- JUL23 |> rename(Jul2023_MaxTemp = Value)
AUG23 <- AUG23 |> rename(Aug2023_MaxTemp = Value)

JUL23 <- JUL23 |> mutate(year = 2023, month = "July", .before = Name)
AUG23 <- AUG23 |> mutate(year = 2023, month = "August", .before = Name)

JUL23 <- JUL23 |> rename(county = Name) |> rename(state = State) |> rename(`July1901-2000Mean` = `1901-2
AUG23 <- AUG23 |> rename(county = Name) |> rename(state = State) |> rename(`Aug1901-2000Mean` = `1901-20

# Column uniformity
H2A$County <- str_to_title(H2A$County)
H2A <- H2A |> rename(county = County) |> rename(state = State)
MHC$county <- str_to_title(MHC$county)
ENV$county <- str_to_title(ENV$county)
AGO$county <- str_to_title(AGO$county)

H2A$state <- str_to_title(H2A$state)
MHC$state <- str_to_title(MHC$state)
ENV$state <- str_to_title(ENV$state)
AGO$state <- str_to_title(AGO$state)
AUG23$state <- str_to_title(AUG23$state)
JUL23$state <- str_to_title(JUL23$state)

# Adding latitude and longitude for mapping
MHC <- MHC |> separate(geolocation, into = c("latitude", "longitude"), sep = ",", convert = TRUE)

# Sum number of migrant health facilities per county
MHC_sum <- MHC |>
  group_by(state, county) |>
  summarise(MigrantHealthCenters = n(), .groups = "drop")


# Binding all data together into one data set to work off of:
JOIN <- left_join(ENV, H2A, by = c("state", "county")) # Join H-2A worker totals and natural disaster d
JOIN <- left_join(JOIN, AGO, by = c("state", "county")) # Join agricultural output data set
JOIN <- left_join(JOIN, MHC_sum, by = c("state", "county")) # Join migrant health center counts
JOIN <- JOIN |>
  left_join(COH, by = c("state", "county")) |>
  left_join(WYH, by = c("state", "county")) |>
  left_join(SDH, by = c("state", "county")) |>
  left_join(NDH, by = c("state", "county")) |>
  left_join(UTH, by = c("state", "county")) |>
  left_join(MTH, by = c("state", "county"))

JOIN <- JOIN |> mutate(MigrantHealthCenters = ifelse(is.na(MigrantHealthCenters), 0, MigrantHealthCente
JOIN <- JOIN |> select(-grep("abbrev", colnames(JOIN), ignore.case = T))

JOIN <- JOIN |> mutate(Hospitals = coalesce(Hospitals.x, Hospitals.y, Hospitals.x.x, Hospitals.y.y, Hosp
JOIN <- JOIN |> select(-Hospitals.x, -Hospitals.y, -Hospitals.x.x, -Hospitals.y.y, -Hospitals.x.x.x, -Ho

JOIN <- JOIN |> mutate(`RuralHealthClinics` = coalesce(`Rural_Clinics`, `Rural Health Clinics`, Rural_He
JOIN <- JOIN |> select(-`Rural_Clinics`, -`Rural Health Clinics`, -Rural_Health_Clinics.x, -Rural_Healt
```

```r
JOIN <- JOIN |> mutate(`CriticalAccessHospitals` = coalesce(`Critical Access Hospitals`, Critical_Access
JOIN <- JOIN |> select(-`Critical Access Hospitals`, -Critical_Access_Hospitals)

JOIN$county[JOIN$county == "lamoure"] <- "la moure"

#JOIN$H2A_workers[is.na(JOIN$H2A_workers)] <- 0


# JOIN <- JOIN |>
#   group_by(state) |>
#   mutate(H2AStateTotal = sum(H2A_workers, na.rm = TRUE))
#-----------------------------------------------------------------------------------------------
JOIN_CUT <- JOIN |>
  select(
    "state",
    "county",
    "Population2020",
    "BuildingValue",
    "AgricultureValue",
    "Areasqmi",
    "H2A_workers",
    "JOIN_CUT$latitude",
    "JOIN_CUT$longitude",
    grep(
      "income|sales|farm_sales|income_net_|commodity_totals|crop_totals_sales|drought|wildfire|heat|hea
      colnames(JOIN),
      ignore.case = TRUE
    ),
    349:length(JOIN)
  )

# Very simplified subset of risk variables
JOIN_CUT <- JOIN_CUT |> select(-grep("Expected|wheat|cattle|hogs|chickens|receipts|historic|index", col
# Filling NAs
JOIN_CUT[sapply(JOIN_CUT, is.numeric)] <- lapply(JOIN_CUT[sapply(JOIN_CUT, is.numeric)], function(x) ife
# Deleting empty columns
JOIN_CUT <- JOIN_CUT[, colSums(!is.na(JOIN_CUT)) > 0]

JOIN_CUT <- JOIN_CUT |>
  left_join(JUL23 |> select(state, county, Jul2023_MaxTemp, `July1901-2000Mean`), by = c("state", "coun
  left_join(AUG23 |> select(state, county, Aug2023_MaxTemp, `Aug1901-2000Mean`), by = c("state", "county

JOIN_CUT <- JOIN_CUT |> arrange(state, county)




cx <- ux <- nx <- mx <- data.frame(state = character(),
                                   correlation = numeric(),
                                   weightedCorrelation = numeric()
)

COLORADO_MHC <- JOIN_CUT |> filter(state=="Colorado") |> arrange(desc(H2A_workers))
```

```r
x <- round(cor(COLORADO_MHC$H2A_workers, COLORADO_MHC$MigrantHealthCenters), 3)
y <- round(cor(COLORADO_MHC$H2A_workers/COLORADO_MHC$Areasqmi, COLORADO_MHC$MigrantHealthCenters/COLORAI
cx <- rbind(cx, data.frame(state = "Colorado", correlation = x, weightedCorrelation = y))

UTAH_MHC <- JOIN_CUT |> filter(state=="Utah") |> arrange(desc(H2A_workers))
x <- round(cor(UTAH_MHC$H2A_workers, UTAH_MHC$MigrantHealthCenters), 3)
y <- round(cor(UTAH_MHC$H2A_workers/UTAH_MHC$Areasqmi, UTAH_MHC$MigrantHealthCenters/UTAH_MHC$Areasqmi)
ux <- rbind(ux, data.frame(state = "Utah", correlation = x, weightedCorrelation = y))

NORTHDAKOTA_MHC <- JOIN_CUT |> filter(state=="North Dakota") |> arrange(desc(H2A_workers))
x <- round(cor(NORTHDAKOTA_MHC$H2A_workers, NORTHDAKOTA_MHC$MigrantHealthCenters), 3)
y <- round(cor(NORTHDAKOTA_MHC$H2A_workers/NORTHDAKOTA_MHC$Areasqmi, NORTHDAKOTA_MHC$MigrantHealthCente
nx <- rbind(nx, data.frame(state = "North Dakota", correlation = x, weightedCorrelation = y))

MONTANA_MHC <- JOIN_CUT |> filter(state=="Montana") |> arrange(desc(H2A_workers))
x <- round(cor(MONTANA_MHC$H2A_workers, MONTANA_MHC$MigrantHealthCenters), 3)
y <- round(cor(MONTANA_MHC$H2A_workers/MONTANA_MHC$Areasqmi, MONTANA_MHC$MigrantHealthCenters/MONTANA_MI
mx <- rbind(mx, data.frame(state = "Montana", correlation = x, weightedCorrelation = y))

state_corr_frame <- rbind(cx, mx, nx, ux)
state_corr_frame <- rbind(state_corr_frame, data.frame(state = "South Dakota", correlation = NA, weighte
state_corr_frame <- rbind(state_corr_frame, data.frame(state = "Wyoming", correlation = NA, weightedCor

colnames(state_corr_frame) <- c("State", "Correlation", "Correlation-Weighted for Sq. Miles")
state_corr_frame <- state_corr_frame |> arrange(desc(Correlation))

# write.csv(state_corr_frame, file = "~/internship/workspace/Written Datasets/H2AWorkers_MHC_WeightedCo


# Combining H-2A population and migrant health center counts
H2AMHC <- H2A |> left_join(MHC_sum, by = c("state", "county"))
H2AMHC$MigrantHealthCenters[is.na(H2AMHC$MigrantHealthCenters)] <- 0 # replace NAs
Z <- JOIN_CUT |> select(state, county, Areasqmi)
H2AMHC <- H2AMHC |> left_join(Z, by = c("state", "county"))

H2AMHC$Areasqmi[H2AMHC$state == "Montana" & H2AMHC$county == "Fremont"] <- 9266
H2AMHC <- na.omit(H2AMHC)

# Calculate correlation
stx <- data.frame(State = character(),
                       Correlation = numeric(),
                       weightedCorrelation = numeric() )

x <- round(cor(H2AMHC$H2A_workers, H2AMHC$MigrantHealthCenters), 3)
y <- round(cor(H2AMHC$H2A_workers/H2AMHC$Areasqmi, H2AMHC$MigrantHealthCenters/H2AMHC$Areasqmi), 3)

stx <- rbind(stx, data.frame(state = "ALL States", correlation = x, weightedCorrelation = y))
colnames(stx) <- colnames(state_corr_frame)
state_corr_frame <- rbind(stx, state_corr_frame)


colnames(state_corr_frame) <- c("State", "Correlation", "Correlation-Weighted for Sq. Miles")
kable(state_corr_frame, align = c("r", "c", "c"))
```

State	Correlation	Correlation-Weighted for Sq. Miles
ALL States	0.289	0.274
North Dakota	0.473	0.486
Montana	0.359	0.258
Colorado	0.325	0.295
Utah	0.196	0.355
South Dakota	NA	NA
Wyoming	NA	NA